Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Lorem Ipsum
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Subtitle

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Principle

The problem

- UPMC Food-101 has been crawled from Google Images
- It contains a certain amount of noise

The idea

- Creating bags from images of 1 class (e.g. pizza)
- Creating bags for "rest" class
- Defining the expected level of noise in the pizza bags
- Using *Learning with Label Proportions* models to detect noise

Experimentations

Protocol

- **Dataset:** $(x_i, y_i, y_i^*) \in \mathbb{R}^p \times \{-1, 1\} \times \{-1, 1\}, \quad i = 1..n$
  - x_i features, y_i noisy label, y_i^* true label (not available for training)
- **Create bags:** Create bags b_j of 30 points having the same y_i and give them a proportion of positive points p_j
- **Training:** Train the SyMIL model on bags and train an SVM on (b_j, p_j) / (x_i, y_i)
- **Evaluation:** Use the decision frontiers of SyMIL / SVM models to reclassify each x_is by predicting $\hat{y}_i^*$, and compare with y_i^*

Some papers for noisy images datasets

Azadi2015 [?]

- Define a regularized loss for training the CNN
- Can be seen as looking for the label of similar images for regularization
- Results slightly better than Sukhbaatar model

References I

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